NYC Taxi Trip Duration Prediction

Project report | Ridge regression (alpha = 1) | target: log1p(trip_duration) | metric: R^2

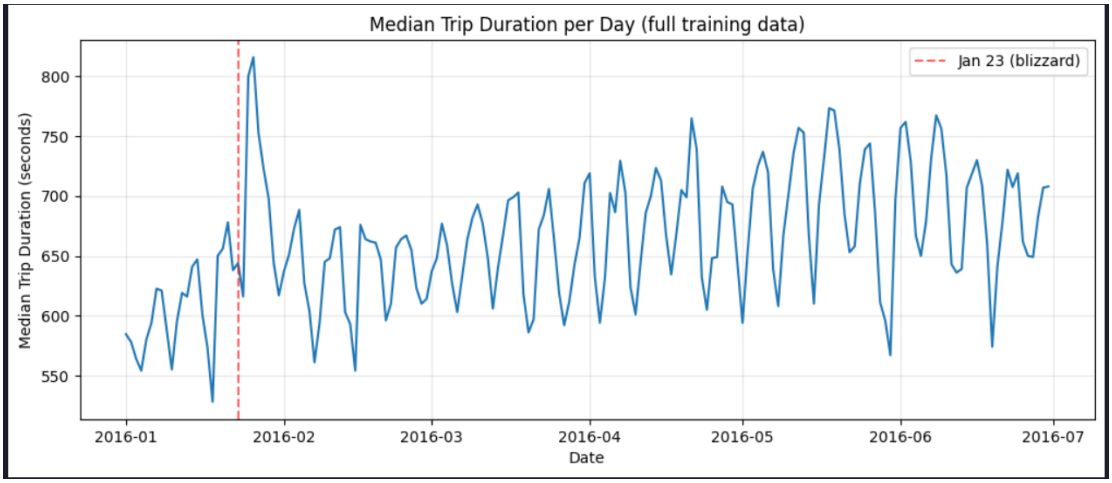
Final model (validation set, 229,319 trips)	R^2 = 0.7107	RMSE = 0.4303 (log scale)
Training set (after cleaning, 994,710 trips)	R^2 = 0.7795	Starting point on the same validation set: 0.6005
Test set (229,322 trips, single evaluation)	**R^2 = 0.7151**	**RMSE = 0.4248 (log scale)**

1. Setup

The model is fixed to Ridge with alpha = 1, so every improvement comes from data cleaning and feature engineering. Exploration was done first on a 5,000-row sample for speed; the main findings were then re-checked on the full training set (1,000,000 trips). The validation set (229,319 trips) was used to compare ideas; the hidden test set is evaluated only once at the very end.

2. Main EDA findings

- **Target.** Durations range from a few seconds to about 24 hours (meter left running), a very heavy right tail. The model is trained on log1p(duration), which is also the scale used for evaluation.
- **Outliers.** Five rules flag impossible trips: duration under 10 s or over 6 h, coordinates outside the NYC box, speed over 100 km/h, and trips longer than 10 min that start and end within 50 m. They flag 5,290 of 1,000,000 trips (0.53%; 0.68% in the sample). Removing them raises the correlation between log distance and log duration from 0.750 to 0.781. Cleaning is applied to the training set only, because the test set will contain such rows too.
- **Distance is the strongest signal.** Correlation with the log target is 0.750 for log distance (full data). Against the raw duration it looked weak (0.185 in the sample) only because of the extreme durations, which is why features must be judged against the log target.
- **Time.** In the sample the median duration ranges from about 498 s (06:00) to about 756 s (15:00). The daily median over the full data (figure below) shows a weekly cycle of about 100 s, a slow rise from January to June, sharp drops on holidays (Jan 18, Feb 15, May 30) and a peak right after the Jan 23 blizzard.
- **Airports.** JFK and LGA trips are 6.6% of trips and differ strongly (table). They are also faster per kilometre than city trips, so distance alone misjudges them.
- **Weak signals.** Raw coordinates are weak linear predictors (|r| about 0.1 in the sample) but carry location information. vendor_id and passenger_count have correlation below 0.03 with the log target; only 36 of 5,000 sample trips have store_and_fwd_flag = Y.



Trip type	Trips (full train)	Median duration	Median distance
JFK	28,382	2,394 s	20.1 km
LGA	36,998	1,636 s	9.6 km
Other	929,330	624 s	2.0 km

3. Features introduced

Feature	Definition	Evidence from the EDA
Log distance	Haversine distance between pickup and dropoff, log1p-transformed	Strongest correlation (0.750); distance is right-skewed
Hour, day of week, day of year	One-hot encoded	Non-linear hourly pattern, weekly cycle, holiday and blizzard effects
JFK / LGA flags	1 if the pickup or dropoff is within 2 km of the airport	Airport trips are 2.6 to 3.8 times longer and faster per km
Zones	KMeans with 500 zones on pickup and dropoff points together; one-hot pickup zone and dropoff zone	Raw coordinates are weak linearly; zones let the model learn area effects
Route	Pickup zone x dropoff zone on a coarser 200-zone map; routes with fewer than 10 training trips share one group	A linear model adds pickup and dropoff effects, but a route has its own effect

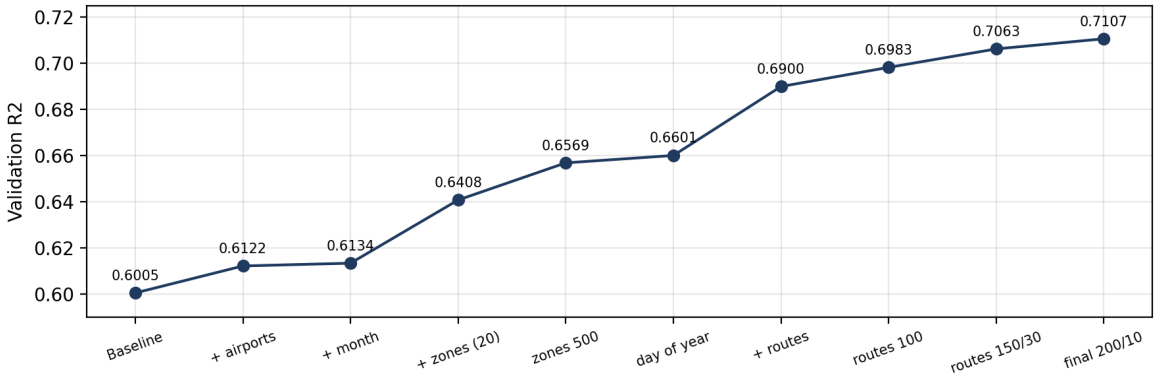
Tried and not kept: month (redundant once day of year is used), distance x rush hour and distance x hour interactions, bearing, Manhattan distance (no reliable gain on the sample), and removing the cleaning step (no change on the validation score).

4. Training

- A single scikit-learn Pipeline: feature engineering, KMeans zones, route id, one-hot encoding and scaling, then Ridge(alpha=1). It accepts raw trip rows, so the saved model (joblib, 4.5 MB) needs no extra preprocessing.
- Cleaning is applied to the training set only; validation and test data are scored as they come. KMeans is fitted on training data only (a random sample of 500,000 points, n_init = 3, random_state = 42), so no validation or test information leaks into the zones.
- Final settings: 500 fine zones; 200 route zones; minimum 10 training trips per route; Ridge alpha = 1.

5. Performance

Step (cumulative, full data)	Val R^2	Val RMSE
Baseline: log distance + hour + day of week	0.6005	0.5057
+ airport flags	0.6122	0.4982
+ month	0.6134	0.4974
+ KMeans zones (20)	0.6408	0.4795
Zones: 500	0.6569	0.4686
Day of year replaces month	0.6601	0.4664
+ route (30 zones, min 100)	0.6900	0.4455
Routes: 100 zones, min 100	0.6983	0.4394
Routes: 150 zones, min 30	0.7063	0.4335
Routes: 200 zones, min 10 (final)	**0.7107**	**0.4303**



- **Test result.** The frozen model was evaluated once on the test set (229,322 trips, not cleaned): R^2 = 0.7151, RMSE = 0.4248. It is slightly higher than the validation score (0.7107), so choosing settings on the validation

set did not inflate the result. No setting was changed after seeing the test score.

- Zone count helped steadily (20 to 500 zones: 0.6408 to 0.6569). Routes gave the largest single gain (+0.030 at first).
- Route settings interact: a high minimum count hurt (200 zones, min 300: 0.6694), while lowering it let more zones help. 300 route zones scored 0.7128, but the train-validation gap grew faster than the gain, so we stopped at 200.
- The validation set contains about 0.5% outliers that are not removed. For an early model, scoring only the non-outlier validation rows raised R^2 from 0.612 to 0.667, so part of the train-validation gap (0.7795 vs 0.7107) comes from those rows rather than from overfitting.

6. Limitations

- Day of year only works for the period covered by the data (January to June 2016) and would not generalize to another year.
- About 25 configurations were compared on the same validation set and the best was kept, so the validation score could be slightly optimistic. The single test evaluation is the unbiased estimate.
- Zones and routes are learned from training trips; locations far from any training trip fall back to the nearest zone, and unseen routes share the rare-routes group.